-Full name: Ha Hoang Phuc
Student ID: 222683
Class: DH22TIN07
LESSON 1: Library Management System
-Identify candidate classes (Book, Reader, Loan, Employee, Publisher)
+ Book: main entity, manages book information.
+ DocGia: book borrower.
+ PhieuMuon: records borrowing/returning.
+ NhanVien: system manager (librarian, storekeeper...).
+ NhaXuatBan: book source information.
- Attributes and behaviors (methods) of each class
+ Item

o Properties: bookID (PK, string), bookTitle (string), author (string), publicationYear (int), publisherID (FK), stockQuantity (int), status (string: "Available available"/"On loan").

o Behavior: themSach (), capNhatSach (), xoaSach (), timKiemSach (), capNhatTonKho ().

+ DocGia

o Attributes: maDocGia (PK, string), hoTen (string), ngaySinh (date), diaChi (string), soDienThoai (string), ngayDangKy (date), gioiTinh (boolean).

o Behaviors: registerCard (), updateInformation (), viewBorrowingHistory ().

+ LoanSlip

o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), employeeId (FK), borrowDate (date), dueDate (date), actualReturnDate (date), status (string: "On loan"/"Returned"/"Overdue").

o Behaviors: createBorrowSlip(), returnBook(), checkOverdue(), calculateFine().

+ Employee

o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), ngaySinh (date), soDienThoai (string).

o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach (). ().

+ NhaXuatBan

o Attributes: maNXB (PK, string), tenNXB (string), diaChi (string), email (string).

o Behavior: themNXB (), capNhatNXB ().

+- Relationships

+Books and Publishers: 1 book – 1 publisher

+Reader and Loan Slip: 1 reader – many loan slips
+Book and Loan Slip: 1 book – many loan slips
+Employee and Loan Slip: 1 employee – many loan slips

EXERCISE 2: Online Sales Management System

- Identify key use cases

Main Actors:

+ KhachHang (Customer / Potential Customer).
+ NguoiQuanLy / Admin (Administrator / Sales Staff).

Key use cases (customer registration, order placement, payment, product management):

KhachHang:

+ DangKyTaiKhoan (Customer registration).
+ DangNhap (Login).
+ XemSanPham (Search / View product list).
+ AddToCart.
+ PlaceOrder.
+ Payment (Online payment / COD).
+ ViewOrders (View order history).

Admin / Manager:

+ ManageProducts (Add/Edit/Delete products).
+ ManageOrders (View/Confirm/Cancel orders).
+ ManageCustomers (Manage customer information).
+ ManagePayments (Process payments, refunds if necessary).

Extend / Include:

+ Payment includes OrderConfirmation.
+PlaceOrder extends Payment (if online payment).

- From use cases, identify necessary classes

+Customer: from registration, viewing orders.
+Product: product management.
+Cart: temporarily stores products before ordering.
+Order: from placing an order, payment.
+OrderDetail: details of products in the order.
+Payment: processes payments (may have subclasses like OnlinePayment, COD).
+Admin or Manager (inherits from User if there's a user system).

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+- Analyze the relationships between classes (specify inheritance if any) +

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+Customer and Order: 1 customer – many orders

+Order and OrderDetail: 1 order – many order details (Composition) +OrderDetails and Product

Pham: many individual details – 1 product

+Orders and Payments: 1 order – 1 payment

+User (abstract class) ←inherits ←Customer and Admin (Customer and Admins are all a type of User, sharing the same account, password, and full name.

Exercise 3: Convert class diagrams to data tables

1. Subject Table

No.	Field name	Data type	Constraint			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	tenMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Course Table

No.	Field name	Data type	Constraint			
			PK	FK	Not NULL	Unique
1	maKhoaHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	